

Data Drift in a Seq2Seq Model for Robot Trajectory Prediction in the RoboCup SSL

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Abstract—Trajectory prediction models are usually reported on a fixed slice of data, which says little about how long they remain useful. This paper audits a sequence-to-sequence (seq2seq) predictor for the RoboCup Small Size League (SSL) that was trained on 2019 world-cup logs and evaluated on the 2021–2025 seasons. The average displacement error (ADE) drifts upward after training, with an excess of up to +0.73 mm in 2022–2025 on top of a +3.05 mm jump in the virtual 2021 edition; the Kalman benchmark degrades even faster, so the neural model’s relative advantage grows under drift. Importance weighting via LSIF attributes 59–72 % of the excess ADE to covariate shift, dominated by tail acceleration; the estimate survives weight clipping and a relative density-ratio transform, and an independent classifier two-sample test confirms the shift in every season. Conservative fine-tuning cuts catastrophic forgetting from 0.82 to 0.46, and a sweep of the elastic weight consolidation penalty over eight orders of magnitude shows that the protection comes from the low learning rate rather than from the Fisher term. Finally, an ADWIN detector calibrated on the error stream under a two-fold holdout by matches reaches a false-alarm rate of 0.04–0.13 per thousand trajectories with 9/10 year-folds covered.

Index Terms—trajectory prediction, data drift, covariate shift, drift detection, RoboCup SSL, mobile robots

I. INTRODUCTION

Predicting where a robot will be a fraction of a second from now is a routine requirement in systems with fast dynamics. In the RoboCup Small Size League (SSL), where six to eleven robots per team move at several meters per second under centralized control, such predictions feed planning, positioning and real-time decision making. [+ o que é a Division A]

The Kalman filter [1] is the traditional benchmark for this task: cheap, deterministic and easy to tune. [+ define o termo Kalman-filter baseline] It also rests on restrictive kinematic assumptions, namely dynamics that are smooth enough to be captured by a local linear model. Sequence-to-sequence (seq2seq) networks relax those assumptions and learn future action patterns directly from logged matches. [+ cita Steuernagel e Máximo aqui] Better accuracy on one season, however, is not the same as being usable three seasons later. Robots are rebuilt, control stacks are rewritten and playing styles change, so a model trained on old logs may quietly decay. That is the data drift problem studied here.

The starting point is the seq2seq predictor of Steuernagel and Máximo [2], which remains the reference architecture for trajectory prediction in the SSL. This paper does not propose

a new architecture. It audits that reference model across seven years of official world-cup logs and answers four questions:

- 1) how the error evolves across seasons, at two prediction horizons, and how that evolution compares to the Kalman benchmark;
- 2) what fraction of the degradation is explained by covariate shift, and which kinematic dimension dominates it;
- 3) whether light adaptation recovers accuracy without catastrophic forgetting;
- 4) whether an online detector can flag the drift with a false-alarm rate low enough to trigger retraining.

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II. RELATED WORK

Trajectory prediction. The field moved from social pooling between neighboring LSTMs (Social-LSTM [3]) to dynamic graphs with multimodal latent variables (Trajectron++ [4]), agent-aware transformers [5] and diffusion models [6]. Within the SSL, Steuernagel and Máximo [2] applied an attention-based seq2seq model at the same horizons used here, and it is the direct baseline of this work. All of these evaluations use fixed data splits; temporal stability across seasons has received far less attention.

Drift and online detection. Gama et al. [7] separate virtual drift, a change in $p(x)$, from real drift, a change in $p(y|x)$. [+ por que a distincao importa na pratica] Detectors are usually error-based, such as ADWIN [8] and Page–Hinkley [9], or data-based, such as KSWIN [10]. Lu et al. [11] argue that calibrating the false-alarm rate is what makes a detector trustworthy in production, and that requirement is taken here as the design constraint of Section VII. [+ EWC é o trade-off estabilidade-plasticidade]

Covariate shift. Direct density-ratio estimation through LSIF or KLIEP [12]–[14] corrects covariate shift without retraining. RuLSIF [15] is the relative-ratio variant with bounded weights, and the classifier two-sample test [16] detects shift without assuming a functional form. LSIF is used here as the main estimator, with the other two serving as robustness checks.

III. DATA AND MODEL

Data. The study uses official logs from the 2019–2025 world cups, excluding 2020, when no in-person championship was held: six Division A matches per year, 36 matches

in total, captured at 60 Hz. The 2021 edition was played entirely in simulation after the pandemic. It is kept in the descriptive analyses because it carries the largest distribution jump of the period (+3.05 mm of excess ADE), but part of that jump plausibly reflects the simulation-to-reality gap rather than organic drift, so it is reported separately throughout and excluded from the adaptation experiments of Section VI. Consolidating a model against an anomalous simulated season would contaminate the old regime with a distribution that no longer describes the league. League composition is constant across the logs, so the observed excess cannot be attributed to a change of division. Per-trajectory analyses sample 8000 windows per match, yielding a temporally ordered stream of 288 000 trajectories, 48 000 per year. The 2019 baseline uses all six matches of the year: a sensitivity sweep shows that a single reference match puts ADE_{2019} anywhere between 4.53 and 6.20 mm, enough to flip the sign of the excess in four of the five remaining years.¹

Model and benchmark. The replicated architecture follows [2]: a 128-unit LSTM encoder, additive Bahdanau attention [17] and an autoregressive LSTM decoder without teacher forcing, trained with Adam [18] at an initial learning rate of 10^{-3} , batch size 1024 and early stopping. Each input step carries six normalized variables: position x, y , velocity v_x, v_y and orientation $\sin \psi, \cos \psi$. [+ como e feita a normalizacao] One detail matters for the drift analysis: the target lives in the space of normalized incremental velocities, and the future position is reconstructed from the last observed state. The model therefore predicts accumulated dynamics, which makes it directly sensitive to changes in the velocity and acceleration distributions. The kinematic comparator is the constant-velocity predictor corresponding to the steady state of a Kalman filter [1]: deterministic, parameter-free and identical in every season, so any degradation on its side reflects robot behavior alone.

Metric. For N trajectories and horizon H , the average displacement error is

$$ADE = \frac{1}{NH} \sum_{i=1}^N \sum_{t=1}^H \|\hat{p}_{i,t} - p_{i,t}\|_2, \quad (1)$$

where $\hat{p}_{i,t}$ is the predicted position and $p_{i,t}$ the observed one. Descriptive results cover the $30 \rightarrow 15$ and $60 \rightarrow 30$ horizons; [+ explica a notacao $a \rightarrow b$] the mechanism and detection analyses use $30 \rightarrow 15$.

IV. TEMPORAL DEGRADATION

Fig. 1 summarizes three descriptive findings.

The seq2seq model degrades outside its training distribution. ADE grows in every post-2019 season, and the excess trends upward through 2022–2025, from +0.23 mm in 2022 to +0.73 mm in 2025 at the $30 \rightarrow 15$ horizon, with 2024 close to zero. The virtual 2021 season stands apart at +67 % for

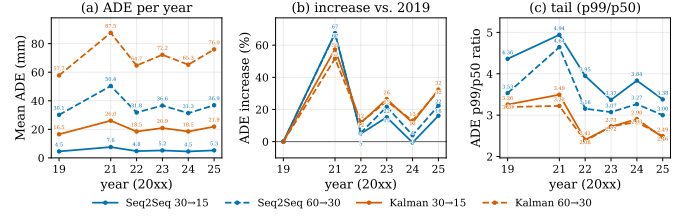


Fig. 1. Temporal degradation (color = model, line style = horizon). (a) Mean ADE per year. (b) Percentage increase in ADE relative to 2019. (c) p99/p50 ratio of the per-trajectory ADE: the 2021 jump is a tail effect ($4.36 \rightarrow 4.94$), whereas in 2022–2025 the ratio falls below baseline while the median rises by about 25 %.

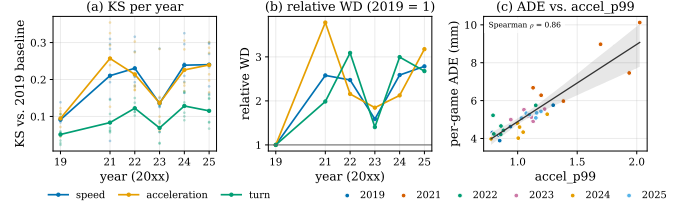


Fig. 2. Kinematic drift. (a) KS statistic per year against the 2019 baseline (lines: annual mean; dots: matches). (b) Wasserstein distance indexed to the 2019 level. (c) Mean ADE per match against the match's $accel_p99$, with a regression line and a 95 % bootstrap band.

the reason given in Section III. The pattern repeats at both horizons.

The Kalman benchmark degrades more. Its relative error growth exceeds that of the seq2seq model in nearly every season. [+ o filtro de Kalman nao e treinado] The right reading is not that the neural model is stable, since its absolute error also grows, but that the kinematic benchmark is even more sensitive to the regime change, so the comparative advantage of learning widens under drift.

The new regime enters through the tail, then spreads. The p99/p50 ratio rises only at the 2021 jump ($4.36 \rightarrow 4.94$), when the model fails disproportionately on a tail of trajectories. In the following seasons the ratio drops below baseline, to 3.4–4.0: the median climbs about 25 % up to 2025 while the 99th percentile stays flat. Behavior that used to be rare has become ordinary, and the gradual degradation now sits in the body of the distribution.

Measuring the input side directly confirms the picture (Fig. 2). Against the 2019 baseline, both the Kolmogorov–Smirnov statistic and the Wasserstein distance show clear and persistent shifts in velocity and acceleration, while turn rate moves much less. At match level, ADE grows almost monotonically with the match's tail acceleration ($accel_p99$, Spearman $\rho = 0.86$): the more aggressive the match, the worse the model does, regardless of the season it belongs to.

V. HOW MUCH OF IT IS COVARIATE SHIFT?

The joint distribution factors as $p(x, y) = p(y|x)p(x)$ [7]. Covariate shift changes only $p(x)$ and can be corrected by reweighting old samples, with no retraining; concept drift

¹Code and processed data will be released upon acceptance; the link is omitted for double-blind review.

TABLE I

IMPORTANCE-WEIGHTING DECOMPOSITION (LSIF) WITH BLOCK BOOTSTRAP BY MATCH ($n_{\text{BOOT}} = 2000$, 6 BLOCKS PER YEAR). BASELINE $\text{ADE}_{2019} = 4.53$ MM; $\text{CORR.} = \text{ADE}_y - \text{ADE}_{\text{IW}}$. 2024 HAS NEGATIVE EXCESS AND IS EXCLUDED FROM THE RECOVERY COLUMN. ALL CLASSIFIER TESTS GIVE $p < 10^{-3}$ BY PERMUTATION. $\dagger$ 2022 HAS A SMALL EXCESS (0.23 MM), SO THE PERCENTAGE SHOULD BE READ WITH CARE.

Year	ADE_y	ADE_{IW}	rec. (%) [95 % CI]	corr. (mm)	AUC
2021	7.58	5.70	61.6 [43.7; 79.3]	1.88 [1.33; 2.42]	0.754
2022	4.75	4.62	59.0 [9.2; 115.2] $\dagger$	0.13 [0.02; 0.26]	0.647
2023	5.21	4.72	71.8 [43.8; 109.2]	0.49 [0.30; 0.75]	0.657
2024	4.50	4.24	—	—	0.612
2025	5.26	4.75	69.3 [57.2; 79.0]	0.51 [0.42; 0.58]	0.710

changes $p(y|x)$ and requires fresh labels. Telling them apart decides whether the cheap fix is available.

LSIF [12]–[14] estimates the density ratio $\hat{w}(x) = p_{\text{target}}(x)/p_{2019}(x)$ directly as a combination of Gaussian kernels, $[+ \text{ define os coeficientes } \alpha]$

$$\min_{\alpha \geq 0} \frac{1}{2} \mathbb{E}_{p_{2019}}[\hat{w}(x)^2] - \mathbb{E}_{p_{\text{target}}}[\hat{w}(x)] + \lambda \|\alpha\|^2, \quad (2)$$

with $x \in \mathbb{R}^8$ collecting the kinematic features of each observed window: mean, p90 and p99 of speed and acceleration, plus mean and p90 of turn rate, clipped at the 99.5th percentile of the pool and passed through $\log(1 + \cdot)$. Stability is monitored by the effective sample size $\text{ESS} = (\sum_i \hat{w}_i)^2 / (n \sum_i \hat{w}_i^2)$, with the usual $\text{ESS} > 0.5$ rule of thumb [14]. The weighted error and the recovered fraction of the excess are

$$\text{ADE}_{\text{IW}} = \frac{\sum_i \hat{w}(x_i) e_i}{\sum_i \hat{w}(x_i)}, \quad \text{rec.} = \frac{\text{ADE}_y - \text{ADE}_{\text{IW}}}{\text{ADE}_y - \text{ADE}_{2019}}. \quad (3)$$

$[+ \text{ define } e_i \text{ e } \text{ADE}_y]$ A recovery of 100 % would mean pure covariate shift, and 0 % pure concept drift.

Robustness. Any quantitative claim here depends on the weights behaving well under the heavy tail of accel_p99 . Three checks support them. First, the weights are bounded: $\max_i \hat{w}_i \in [2.9; 4.4]$ in every season, and the ESS of 0.58–0.73 comes from the near-zero mass assigned to the new regime, which is the correct mechanism for reconstructing 2019, not from extreme weights (the top 1 % of trajectories carries only 3–4 % of the mass). Second, capping the weights at $c \in \{1.5, \dots, 50\}$ leaves recovery saturated from $c \approx 3$ onward. Third, applying the RuLSIF relative-ratio transform $\hat{w}/(\alpha \hat{w} + 1 - \alpha)$ with $\alpha = 0.1$ keeps recovery at 59–65 %. Fitting RuLSIF directly was tested and discarded: in this configuration it returns nearly uniform weights ($\text{ESS} \geq 0.94$) and underestimates recovery at 9–23 %, an artifact of the fit rather than of the estimand, as the two previous checks show.

Reading. Table I places LSIF recovery at 59–72 % in the years with a positive excess, with confidence intervals excluding zero. The stronger statement is on the absolute correction, whose lower bound stays positive even in 2022 (≥ 0.02 mm): reweighting moves the error toward the 2019 level by an amount distinguishable from zero with only six bootstrap blocks. The classifier two-sample test agrees, separating each

TABLE II

MARGINAL RECOVERY PER FEATURE (LSIF REFITTED ON ONE FEATURE AT A TIME), FOR THE THREE SEASONS WITH A SIGNIFICANT EXCESS.

Feature	2021 (%)	2023 (%)	2025 (%)	mean ESS
accel_p99	57.1	52.6	51.7	0.55
accel_p90	54.0	51.3	51.1	0.57
accel_mean	43.0	47.0	49.5	0.68
speed_p99	17.2	36.1	40.1	0.84
speed_p90	15.8	34.4	38.0	0.85
speed_mean	7.9	24.7	27.4	0.91
turn_p90	3.9	0.2	0.0	0.97
turn_mean	5.3	0.9	0.0	0.96

TABLE III

ADAPTATION STRATEGIES UNDER THE COMMON CONSERVATIVE PROTOCOL. THE UNADAPTED MODEL SCORES 4.584 MM ON THE OLD REGIME AND 4.965 MM ON THE NEW DATA. EWC REPORTS THE MEDIAN OF THE $\lambda \in [50; 10^7]$ PLATEAU. THE LAST THREE ROWS ARE DRIVEN BY THE COVARIATE DIAGNOSIS.

Strategy	cat_ratio $\downarrow$	ADE (mm)	
		old	new
Conservative FT ($r = 0$)	0.457	4.480	4.762
EWC (λ plateau)	0.457	4.481	4.817
Uniform replay ($r = 0.5$)	0.438	4.439	4.760
Importance-weighted replay	0.417	4.404	4.761
Acceleration-selective FT	0.467	4.526	4.811
Selective + IW replay	0.449	4.441	4.800

season from 2019 with $\text{AUC} \in [0.61; 0.75]$ and ranking the seasons by severity in the same order, without estimating any density ratio. The unrecovered 30–40 % is residual concept drift plus sampling noise.

Refitting LSIF on one feature at a time makes the mechanism concrete (Table II). Tail acceleration leads consistently, followed closely by accel_p90 ; speed features recover far less, and turn-rate features are essentially inert, below 5 % in every season. The ranking is stable across seasons: the league has adopted more aggressive acceleration profiles, and that is the main vector of degradation. The ESS column moves the other way, as it should: the features that shifted the most are the ones whose reweighting costs the most effective sample.

VI. ADAPTATION

Since the excess is mostly covariate, four families of adaptation strategies were compared under one conservative protocol: old regime = 2019, new data = 2022–2025 (500 trajectories per match, 2021 excluded as an outlier), frozen encoder, two trainable decoder layers, $\text{lr} = 10^{-5}$ and five epochs. Forgetting is measured by cat_ratio , the fraction of old-regime trajectories that get worse after the update. Table III collects the results.

Naive fine-tuning is destructive: without the conservative protocol, cat_ratio reaches 0.82 after a single epoch. Elastic weight consolidation [19] is the standard remedy, adding $\frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2$ with F_i the diagonal Fisher information. The finding here is that the Fisher term does nothing: sweeping $\lambda \in \{0, 50, \dots, 10^7\}$ leaves cat_ratio pinned near 0.46

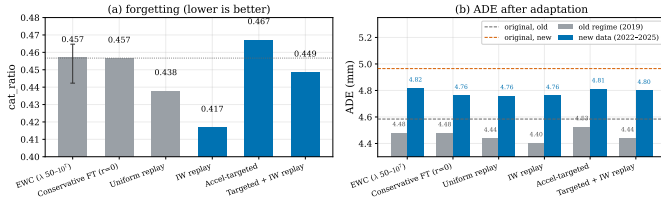


Fig. 3. Adaptation strategies under the same protocol (blue: variants driven by the covariate/acceleration diagnosis). (a) Forgetting, with the error bar showing the EWC λ plateau and the dotted line the no-replay control. (b) ADE after adaptation on both regimes, with the unadapted model as reference (dashed lines).

across eight orders of magnitude, because the diagonal is numerically tiny ($F_i \approx 10^{-7}$), as expected from an already converged decoder. What protects the old regime is the small learning rate and the frozen encoder.

Mixing a replay buffer of original data into the update, at a replay-to-new ratio $r = 0.5$, improves all three metrics at once, pushing `cat_ratio` below the no-replay control and reaching the lowest ADE on new data in the table. The gain is modest, since the conservative schedule already limits forgetting, but it is consistent and essentially free.

Steering adaptation by the covariate diagnosis gives a more interesting split (Fig. 3). Sampling the replay buffer proportionally to $\hat{w}(x)$ helps: it reaches the lowest `cat_ratio` (0.417) and the lowest old-regime ADE in the table, though by a small margin on a single seed. Hard selection hurts. Fine-tuning restricted to the top 50 % of new data by tail acceleration, exactly the region where $p(x)$ moved, produces the worst forgetting of all six rows (0.467), above even the control. The tail is the subset farthest from the 2019 regime, so training only on it displaces the weights more; the unchanged body of the distribution, which hard selection throws away, acts as an implicit anchor. Reweighting preserves that anchor and merely emphasizes the relevant tail. The covariate diagnosis is therefore useful for monitoring, for explaining the excess and for weighting replay, but not for discarding data.

VII. ONLINE DETECTION

The operational question is whether drift can be flagged automatically on the 288 k-trajectory stream. A detector is judged by three quantities [11]: FAR_{2019} , the alarms per thousand trajectories in the training year, where no drift exists [+ e todo alarm e falso] (admissibility ceiling 0.20/1k); coverage, the number of 2021–2025 seasons with at least one alarm; and latency after each year boundary. To avoid tuning and reporting on the same data, calibration runs under a two-fold cross-validation by matches: half the matches of each year select the ADWIN [8] configuration out of a 96-point grid, and the untouched half is evaluated once. All figures below are out of sample.

Local robust normalization of the per-trajectory signal, by median and MAD over a 200-trajectory window, is what makes the detector viable at all. On the raw signal no configuration in the grid is admissible in either fold, and the 21 conventional

TABLE IV
LAYERED DETECTION ARCHITECTURE, WITH METRICS ON THE TEST SPLIT OF EACH FOLD (A/B). ALL SIGNALS USE ROBUST NORMALIZATION WITH $w = 200$; EACH DETECTOR IS CONFIGURED PER FOLD DURING CALIBRATION.

Layer	Signal	FAR/1k (A/B)	Cov.	Action
Primary	seq2seq ADE	0.042 / 0.125	5/5, 4/5	retrain
Watch	accel_p95	0.292 / 0.250	5/5, 5/5	monitor
Confirmation	AND(ADE, accel)	0.000 / 0.000	3/5, 2/5	confirm
Corroborator	Kalman ADE	0.167 / 0.083	5/5, 5/5	redundancy

smoothers tested (moving averages, EWMA, Holt, Savitzky–Golay) all increase FAR by injecting the autocorrelation that ADWIN assumes away. On the normalized ADE the winning family is stable across folds ($\delta = 10^{-4}$, windows 1000/5000), giving FAR_{2019} of 0.042/1k with 5/5 coverage in fold A and 0.125/1k with 4/5 in fold B, at a latency of 0.2 to 1.5 matches. The holdout does more than add confidence here; it changes what one would deploy. The configuration that in-sample selection picks keeps $\text{FAR} \approx 0$ and then covers only 2–3 of 5 seasons out of sample.

The diagnosis of Section V also buys a second monitoring channel. Because the shift is dominated by tail acceleration, the features of the observed window carry the drift signal without running the model and without waiting for the 15 future steps that ADE requires. Applying the same protocol to four kinematic signals reproduces the ranking of the per-feature decomposition, with `accel_p95` ahead ($\text{FAR} = 0.25\text{--}0.29/1\text{k}$, 5/5 coverage in both folds).² No single feature is admissible on its own, since kinematics vary naturally within 2019, but the acceleration channel fires before the error channel in 8 of 10 year-folds, which is enough for a watch layer. At the other extreme, the conjunction of the two alarms within a coincidence window of 200 trajectories drives observed false alarms to zero in both folds, at the cost of coverage (2–3 of 5).

Table IV summarizes the resulting three-layer architecture. Two caveats and one negative result deserve mention. Per-trajectory FAR assumes approximately independent trajectories within 2019, and the serial correlation is real ($\rho_1 = 0.54$); a permutation test shows the violation is conservative, since shuffling trajectories within matches drives false alarms to zero in all 200 replicates. Among alternatives, Page–Hinkley [9] aggregated over windows achieves $\text{FAR} = 0$ but only detects the abrupt 2021 jump, and KSWIN [10] is inadmissible in every configuration ($\text{FAR} \geq 0.438/1\text{k}$). The negative result is structural: combining detectors that run at different time scales is unworkable, because the closest pair of alarms is 214 trajectories apart, beyond any usable coincidence window. Aligning the scales fixes it, which is how the confirmation layer above was obtained.

²p95 rather than the dominant p99, because the observed window holds only 29 acceleration samples.

VIII. CONCLUSION

A seq2seq trajectory predictor for the SSL loses accuracy as the league moves away from its training season. The loss is gradual, it is measurable, and most of it can be traced to a specific change in the input distribution. Excess ADE trends upward through 2022–2025 on top of the jump in the virtual 2021 edition, while the Kalman benchmark degrades faster still. Between 59 and 72 % of that excess is recovered by importance weighting, an estimate robust to weight clipping and to the relative-ratio transform and corroborated by an independent classifier test, with tail acceleration as the dominant vector.

On the response side, light adaptation keeps forgetting under control (cat_ratio from 0.82 to 0.46), EWC turns out to be inert over eight orders of magnitude of λ , and a replay buffer improves forgetting and adaptation at the same time, with the optimum at importance-weighted replay. Hard selection by the drifting tail backfires. On the operational side, ADWIN over a robustly normalized error stream reaches a false-alarm rate of 0.04–0.13 per thousand trajectories out of sample with 9/10 year-folds covered, and the same diagnosis yields a model-free acceleration channel that anticipates the error alarm in 8 of 10 year-folds.

Relative to [2], the contribution is a shift of question: not whether the seq2seq model beats the kinematic benchmark on a fixed split, but how that advantage evolves as the distribution drifts away from the training season, and what can be done about it.

Several limitations bound these conclusions. Only Division A appears in the logs. Six matches per season keep the block confidence intervals wide, and the Wilson intervals of the holdout cross the admissibility ceiling, so claiming admissibility at 95 % confidence would require more 2019 matches. ADE is single-mode and would have to give way to min-ADE_K [4], [5] for multimodal predictors. Robust normalization assumes gradual drift and would absorb part of an abrupt change. The residual concept-drift fraction is not isolated per feature and may hide covariate shift in unmeasured dimensions. Finally, the 2021 season was played in simulation, so its error jump mixes the simulation-to-reality gap with seasonal drift, and the gradual-trend conclusions rest on 2022–2025. Deeper adaptation, retraining the encoder with an importance-weighted replay buffer, is the natural next step.

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